EAST WEST UNIVERSITY

Lab 6

Text Mining with YouTube Data

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Introduction

This week's lab o ***"Text Mining with YouTube Data"*** is a part of the previous cleaning and preprocessing steps that were performed in the second lab. Previously, text data were principally characterized by examining the mining of such patterns as slight changes, and clustering, while this lab is inclined to deriving insights from comments and captions that are founded upon linguistic rules. The main idea of this lab is to evaluate the level of similarity and dissimilarity between the video content (captions) and the audience's opinion in the comment section. Since the lab is going to apply TF-IDF keyword extraction, bigram analysis, and sentiment classification, it will then move on to discussing how themes come, overlap, or diverge across the two datasets.

The analysis is a very crucial part that proves how different the content published by the video maker (captions) is from that of the audience (comments). The text mining result that structure, topics, and emotional tone are revealed across different but related textual data sources is by and large the overall workflow showing.

Objectives

- Identifying the most important keywords using TF–IDF for both datasets.
- Comparing overlapping and unique themes using a Venn diagram.
- Analyzing commonly occurring bigrams to capture short phrase patterns.
- Appling sentiment analysis to measure emotional tone in comments versus captions.
- Connecting results with earlier findings from Labs 3, 4, and 5 to understand how different text-mining approaches complement each other.

Tools and Libraries

- **Python 3.x** - for executing codes on Google Colab.
- **pandas & numpy** – for importing the dataset and managing the DataFrame objects.
- **matplotlib** – for importing the dataset and managing the DataFrame objects.
- **matplotlib-venn** – for drawing Venn diagram.
- **scikit-learn** – for vectorization by TF-IDF.
- **nltk (VADER)** – for sentiment classification.

- **itertools** – for finding co-occurring keyword pairs
- **google.colab.files** – for the transfer of files between notebook and local storage.

All development was done in **Google Colab**, which provided a simple environment for testing and visualizing the clustering steps.

Dataset Description

The lab is working with two datasets that were previously cleaned up and created in 2nd lab.

- cleaned_comments.csv
- cleaned_captions.csv

The two datasets have the same cleaned_tokens column that has the words that have been normalized and tokenized. The cleaning removed unnecessary characters, emojis, repeated symbols, and other noise that had been mixed up in the text and made it writable and readable.

The captions dataset gives the same content as the YouTube video transcribed. It is more coherent, grammatical, and instructional. The comments dataset is more spontaneous and closer to the users' feelings and moods, thus, being more diverse, emotional, and informal.

Prior to analysis, token lists were concatenated into a single text string for TF–IDF and sentiment tools to work correctly. The combination of these two datasets allowed for a direct juxtaposition of how a video transmits information and how the audience reacts to it.

Methodology

1. Data Loading & Preparation

A brief examination verified that all rows had a token list appropriate for the upcoming analysis. F-IDF needs uninterrupted writing, so the tokens were combined by using space separation to create a new text column for each dataset. This assured that the input for keyword, bigram, and sentiment extraction was uniform. After that, the datasets were placed into a working folder, and all outputs were routed there for uncomplicated organization.

2. TF–IDF Keyword

TF–IDF was used on comments and captions to find which words were the most significant for each text type. The model was rather easy-going **`min_df=2`** to overlook uncommon words, **`max_df=0.85`** to take away very common ones, and **`ngram_range=(1,1)`** to work with single words only. The top 15 TF–IDF keywords for both datasets were written to CSV files, and bar charts were displayed.

3. Keyword & Theme Comparison

The top 20 TF–IDF keywords obtained through comments and captions were turned into sets for an analysis of thematic overlapping. As a result, three classes were marked: common keywords, keywords exclusive to comments, and keywords exclusive to captions. A Venn diagram illustrated these differences very clearly. The captions were largely composed of the terms that describe and instruct, while the comments expressed more personal reactions and informal expressions.

4. N-gram (Bigram) Analysis

A TF–IDF model with **`ngram_range=(2,2)`** was applied to both datasets for extracting bigrams. The ten most important bigrams were stored in CSV files, providing insights at the level of short phrases. Bigrams from captions were much more meaningful because they were composed of complete sentences. Whereas comments, due to their informal style and concise nature, produced less structured phrases. This was consistent with Lab 5's finding that comment TF–IDF vectors were sparse and inconsistent.

5. Sentiment Analysis

VADER was used for sentiment analysis. Each comment and caption was labeled as positive, neutral, or negative, and bar chat was used to display the segments. This connects with Lab 4’s finding that user reactions shift over time, whereas captions contain stable information. Summary CSVs were generated to store sentiment counts and percentages.

6. Linking to Previous Labs

- **Lab 3 (Co-occurrence):** The captions keywords were related to teaching, which were very much like the robust itemsets found in Lab 3. The comments continued to be disorganized and casual, reflecting the loose arrangement observed earlier.

- **Lab 4 (Temporal Analysis):** In Lab 4 it was observed that the comments gradually gained importance during the study, and they were increasingly distant from the data and fragments of the link. The trend was consistent with the findings of the sentiment analysis in Lab 6.

- **Lab 5 (Clustering):** The three themes of Lab 5, positive engagement, technical/API terms, and URL fragments, were represented by the same TF-IDF keywords in Lab 6, proving thereby the same consistent patterns across the clustering and keyword extraction methods used.

Results and Discussion

1. TF–IDF Top Keywords

Top 15 TF-IDF keywords (Figure 1 and 2) indicated that the commentary and the caption are markedly different from each other in terms of vocabulary.

a. Comments

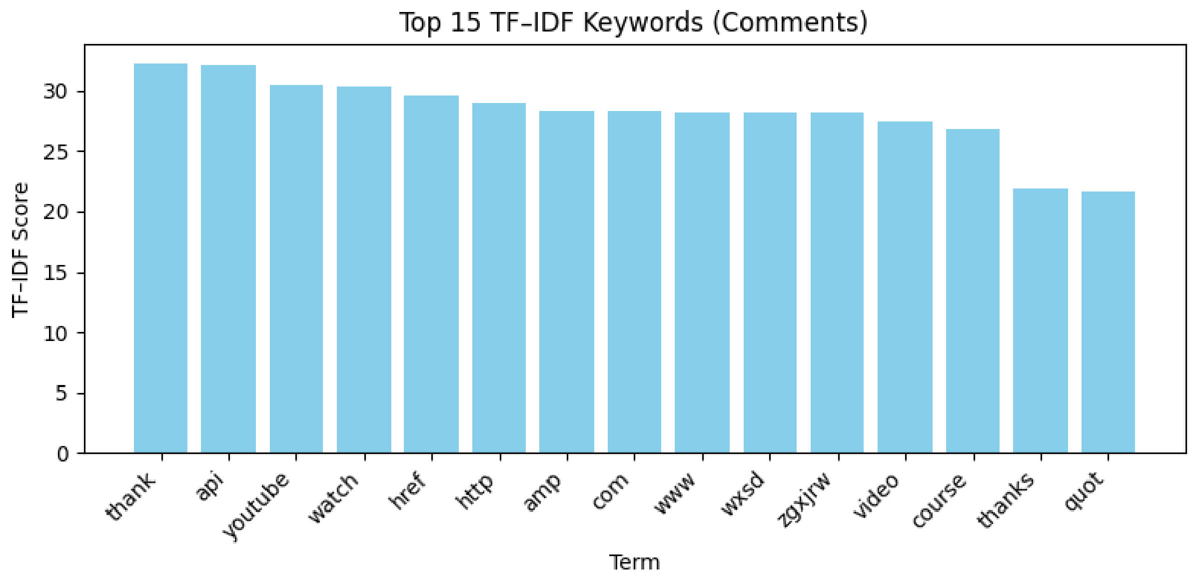

Figure 1. Top 15 Keywords (Comments)

These keywords reflect:

- URL fragments (**http**, **www**, **amp**, **href**)
- Viewer reactions (**thank**, **video**)
- Technical/API references (**api**)

b. Captions

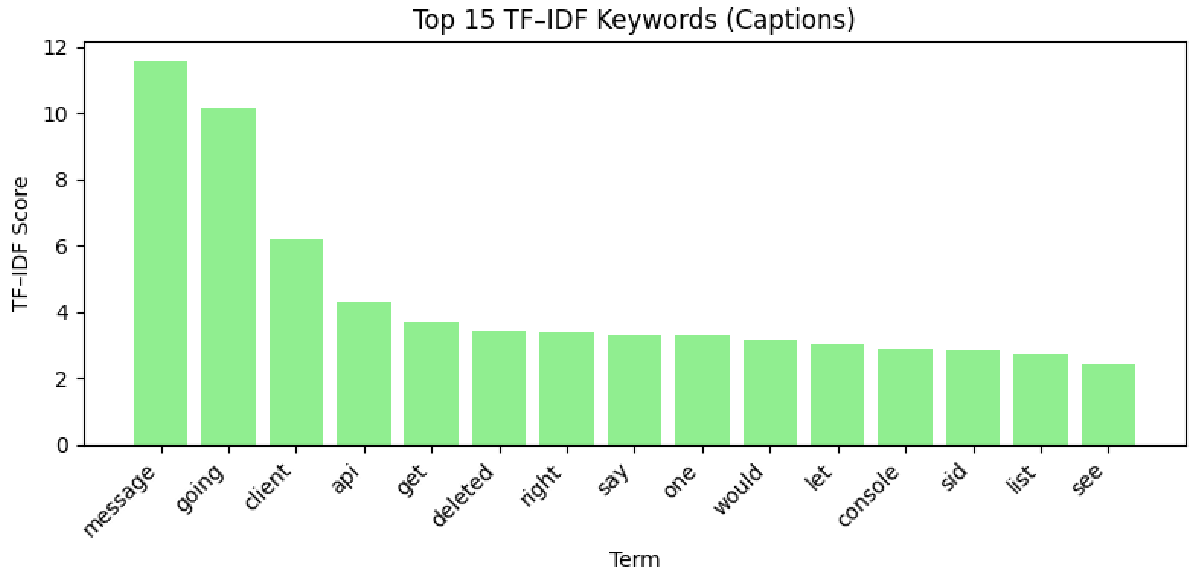

Figure 2. Top 15 Keywords (Captions)

These are more structured and instructional, consistent reflecting creator's explanation style.

c. Discussion

It can be said comments are mostly links, reactions, and platform-specific tokens, whereas captions which highlight instructional terms, commands, and step-by-step descriptions. The distinction between audience-generated and creator-generated texts is thus underlined.

2. Keyword Overlap (Venn Diagram)

- **Comments-only (17 terms):** video, youtube, great, http, href, token, spotify, amp, like, course, thanks
- **Captions-only (17 terms):** sid, going, want, deleted, function, list, message, client, console, let
- **Common (3 terms):** get, api, com

There is an extremely small area of overlap between the two datasets. The captions are more oriented towards the task, while the comments show reactions and include metadata tokens. The little common vocabulary is restricted to very generic technical terms.

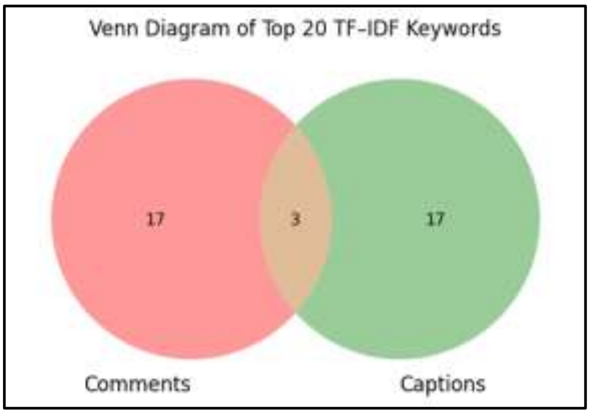

Figure 3. Venn Diagram

3. Top Bigrams

Dataset	Top Bigrams (Unique)
Comments	zgxjrw amp, wxsd zgxjrw, www youtube, href http, com watch, api course
Captions	message client, going say, would deleted, deleted message, going get, error console, uri going, say console

Table 1. Unique Bigrams in Comments vs Captions

- Captions contain **meaningful, action-oriented** indicating the instructional nature.
- Comments contain **URL fragments, metadata,** matching the noise patterns seen in Lab 3 and Lab 5 observations earlier.
- There are **no common bigrams**, confirming difference of language between captions and comments.

4. Sentiment Distribution

Comments (963) and Captions (62) both were labeled **100% neutral**.

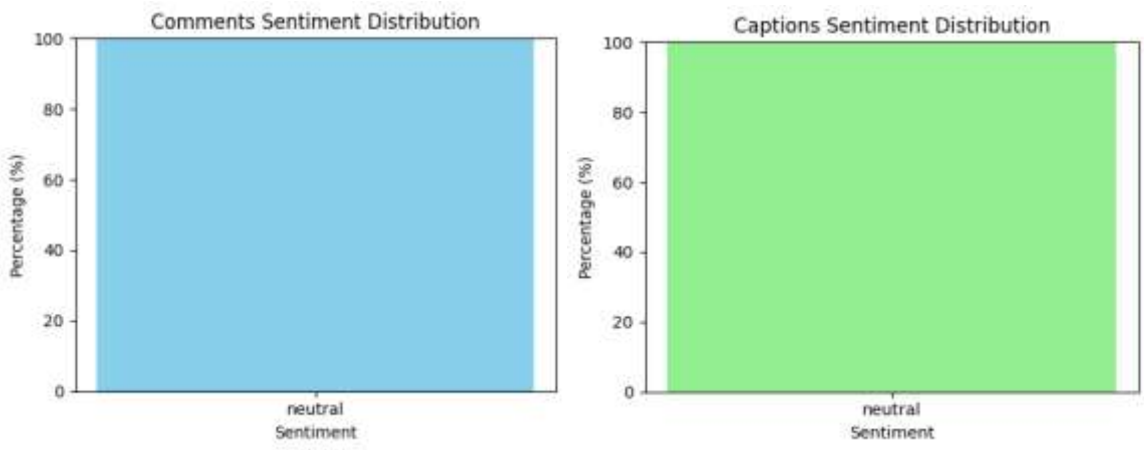

Figure 3. Sentiment Analysis

It can be said that

- Comment section lacks no emotional words after cleaning (because the dataset contains many technical or URL entries).
- Captions naturally stay neutral because they are instructional purely.

5. Co-Occurrence Analysis

term1	term2	count
com	watch	121
amp	com	120
com	youtube	119
href	http	117
com	http	117

Table 2. Unique Bigrams in Comments vs Captions

By observing the co-occurrences, the claim that comments are heavily dominated by **URL fragments** is strongly proven.

Critical Prompts Answers

1. What themes did you expect to find before running TF–IDF?

My initial idea was to find two types of content, technical and instructional. One term used in the captions and the other informal one reaction or replies of the viewers in the comments.

2. Which dataset showed more diverse language? Why?

The captions were more linguistically diverse in terms of meaning. They contained whole sentences that described the steps and the concepts of the topic API. Comments consisted of metadata tokens and repeated URL fragments, which decreased their linguistic diversity.

3. What do bigrams reveal that single keywords do not?

Bigrams show **short phrase structures** highlighting relationships between words. This reveals instructions that single-word keywords cannot capture.

4. How do sentiment differences indicate audience reactions vs creator intentions?

Both csvs were neutral. However, captions are neutral by design as they provide instructions. On the other hand, comments appeared neutral because the dataset is dominated by link-styled tokens rather than emotional reactions. Therefore, captions reflect creator intention, while comments revealed audience emotional engagement.

Conclusion

The lab marked a comparison between the YouTube captions and comments in terms of language, structure, and sentiment. The TF–IDF method indicated that captions were more orderly and educational, whereas comments were characterized by links, metadata fragments, and brief reactions. The Venn diagram and bigram analysis still illustrated that the two datasets had very little in common when it came to meaningful phrases. The sentiment analysis classified both datasets as neutral, mainly because captions were purely informational and the comments contained mostly non-emotional tokens. In general, the analysis has demonstrated a distinct separation between the content generated by creators (captions) and that generated by the audience (comments). Captions give directions and perform the actions while the comments in this dataset are just technical noise or repeating patterns. These revelations are in good agreement with earlier labs and are confirmation of the fact that different text-mining techniques are revealing the same structure and quality of YouTube text data consistently.